

Material Texture Design: Texture Representation System Utilizing Pseudo-Attraction Force Sensation

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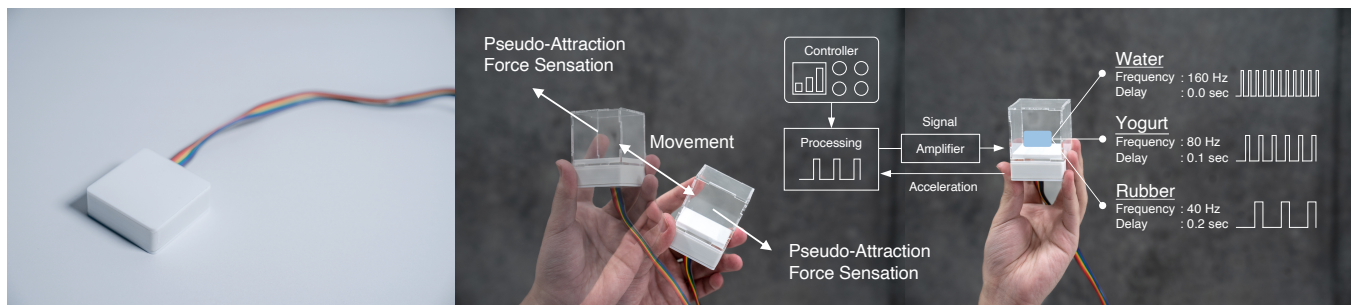


Figure 1: Appearance of device and the mechanism for material texture design

ABSTRACT

We propose Material Texture Design, a material texture representation system. This system presents a pseudo-attraction force sensation in response to the user's motion, and displays a shear sensation at the fingertips. The user perceives a change in the center of gravity from the shear sensation and feels the artificial material texture. Experimental results showed that the perceived texture could be changed by adjusting the frequency. Through demonstration, users can distinguish different textures such as water, jelly, or a rubber ball, depending on the frequency and latency. We propose this system as a small, lightweight, and simple implementation system for texture representation.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

texture design, haptic display, elastic interface

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1 INTRODUCTION

Conventional haptic interfaces are inorganic, inert, and immature for representing the richness of material texture, and this can be a bottleneck in rendering the virtual world to the physical experience. Therefore, we propose Material Texture Design, a system for representing the texture of materials (Fig. 1). The system incorporates an acceleration sensor to capture the user's motion and presents a pseudo-attraction force sensation [Rekimoto 2013] at the fingertips in response to the acceleration data. Here, the force sensation was constructed using a small and powerful haptic actuator. The force sensation induces a perceptual change in the center of gravity of an object, and the perceptual change creates the sensation that the material is dynamically moving in response to the user's motion. Due to this effect, the user perceives the characteristics of the material, such as a feeling of jiggly or viscosity.

The particular feature of this system is the use of a three-fold illusion. The first illusion is the vivid texture perception or the simple device. Here, the perception of the center of gravity is just changing. The second illusion is the change in the center of gravity. In reality, it does not change; only a shearing sensation is presented at the fingertips. The third illusion is shear perception. This is constructed through a pseudo-attraction force sensation, where actual force is not presented. This mechanism allows the presentation of dynamic material texture while keeping the advantages of small size and lightweight. In our previous work ([Hirose and Inami 2021]), we also constructed multiple illusion mechanisms that utilized changes in the center of gravity, in this case as well.

Another key feature is the ability to reproduce various textures with only one signal for pseudo-attraction. Compared to the other methods used to represent material texture in haptic research [Tanaka et al. 2020], this is an extremely simple implementation. This is due to the fact that pseudo-attraction is a powerful method and the interactive design of the system, which is presented in accordance with human actions, works well for perception.

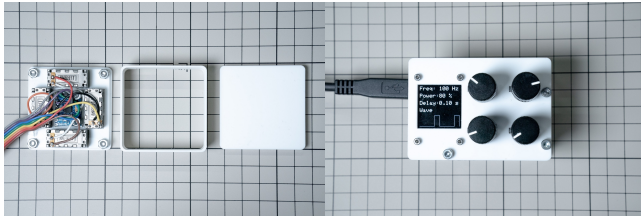


Figure 2: Left: Controller, Right: Inside the module

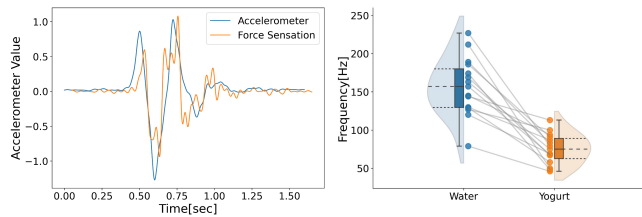


Figure 3: Left: Force sensation response to acceleration, Right: Frequency perception of water and yogurt

This paper describes the system design, experimental results, and comments on the demonstration. We propose it as a new simple texture representation system and material texture design method.

2 IMPLIMENTATION

The system consists of four components, a module, an amplifier [Minamizawa et al. 2012], a Raspberry Pi 4, and a controller. The inside of the module is shown in Fig. 2 (left). An accelerometer (MPU-6050) and four haptic actuators (AFT14A903A) are mounted. These actuators provide forces in the biaxial direction. The module is 49 mm long, 49 mm wide, 18 mm high, and 52 g weight. The module is connected to a Raspberry pi 4 and an amplifier to amplify the vibration. The controller can control the main parameters of the texture: (1) frequency, (2) power, (3) latency, and (4) waveform. Fig. 2 (right) is an example of signal control. We provide it as a platform for creating various textures.

3 EXPERIMENT

For system validation, we performed two experiments. First, we measured the force sensation in response to acceleration. The results in Fig. 3 (left) showed that the minimum system latency was 50 ms. Second, we conducted an experiment to clarify the relationship between the frequency of force sensation and material texture perception. The experiment was designed using an adjustment method. 15 subjects adjusted the frequency and force so that they felt the same texture as "water" or "yogurt". Fig. 3 (right) is the result. It was found that a low-viscosity texture like water could be expressed at high frequencies, while a thick texture like yogurt could be expressed at low frequencies. The variance of the frequency of water is large, while that of yogurt is small. The results indicate that all participants responded that the perception of yogurt is closer to a low frequency than that of water.

4 DEMONSTRATION

We demonstrated it to 30 people and collected comments. All participants could perceive the presence of the material texture change



Figure 4: Users in the demonstration.

(Fig. 4). Most of the comments were about the textures. Participants perceived water, slime, rubber balls, jelly, yogurt, sand, mercury, etc., and the perception of viscosity and amount changed with frequency and latency. The following comments indicate that the force sensation design worked well;

I feel inertia insanely..... It's overshooting. It feels like something is jumping out of my hand.

Some have commented on perceptual changes with latency;

Increasing the latency increases the viscosity. With a latency setting of about 1000 ms, when tilted, I feel as if the heavy object is moving slowly.

The force sensation likely represents the global motion of the material, while the frequencies of the waveforms composing the force sensation represent the local motion of the material.

5 CONCLUSION

We proposed Material Texture Design, a system to represent the material texture using pseudo-attraction force sensation. In this paper, we described the system design and the experimental results. The experimental results showed that the system could express texture changes depending on frequency. 30 people experienced the system through the demonstration, and they could feel various textures such as water, jelly, and rubber ball by changing the parameters (frequency, latency, power, and waveform). At SIGGRAPH, we plan to conduct a perception questionnaire through the demonstration.

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